Coding & Solution

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection Learning Support Engine
Maximum Marks	5 Marks

Coding & Solution

This document evaluates the quality and completeness of the implemented Emotion Detection Learning Support Engine. The solution is developed using Python and Streamlit with BiLSTM, BERT, and Gemini AI to detect student emotions and provide personalized learning recommendations. The project follows a modular architecture, uses meaningful variable names, includes comments for better readability, and integrates reusable components such as authentication, analytics, report generation, and session history to ensure maintainability and scalability.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/akshayashiny/Emotion-Detection-Learning-Support-Engine
Programming Language(s)	Python

Framework(s) Used	Streamlit, TensorFlow, PyTorch (Transformers), SQLite
Key Features Implemented	• User Authentication • Emotion Detection using BiLSTM • Emotion Detection using BERT • AI-powered Learning Support using Gemini AI • Personalized Learning Recommendations • Session History Management • Analytics Dashboard • Downloadable Analysis Report
Pending / Incomplete Features	No major pending features.
Setup / Run Instructions	1. Install the required Python packages using requirements.txt. 2. Run the application using: streamlit run app.py 3. Open the local URL generated by Streamlit in a web browser. 1. Install the required Python packages using requirements.txt. 2. Run the application using: streamlit run app.py 3. Open the local URL generated by Streamlit in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes

6	Code is committed to a version control repository	yes

Additional Notes / Comments:

The project follows a modular architecture with reusable components for authentication, emotion detection, recommendation generation, analytics, session history, and report generation. The code is organized for readability, maintainability, and future enhancements.